

BCAP-Free Injectivity of the Calkin Representation

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Status

This packet records a candidate partial result, likely valid. It concerns Remark 1 after Theorem 2.1 of Boedihardjo–Johnson, arXiv:1301.0054, which asks whether their Calkin representation theorem remains true without the hypothesis that X has the bounded compact approximation property (BCAP).

Source Context

Boedihardjo and Johnson define, for an infinite-dimensional Banach space X , an ultrapower space $\widehat{X}$ and for $T \in B(X, Y)$ an induced operator $\widehat{T}$. Compact operators satisfy $\widehat{K} = 0$. Their Theorem 2.1 states that if X has the λ -BCAP, then

$$f : B(X)/K(X) \rightarrow B(\widehat{X}), \quad \dot{T} \mapsto \widehat{T},$$

is a norm-one $(\lambda + 1)$ -embedding.

Let X be an arbitrary infinite dimensional Banach space and let $(\mathcal{U}, \Lambda_0)$ be the ultrapower (see e.g., [2, Chapter 8]) of Y^* with respect to U . (The ultrafilter U and the directed set Λ_0 do not depend on Y .) If $(y_\alpha^*)_{\alpha \in \Lambda_0}$ is a bounded net in Y^* , then its image in $(Y^*)^U$ is denoted by $(y_\alpha^*)_{\alpha, U}$. Consider the (complemented) subspace

$$\widehat{Y} := \left\{ (y_\alpha^*)_{\alpha, U} \in (Y^*)^U : w^*\text{-}\lim_{\alpha, U} y_\alpha^* = 0 \right\}$$

of $(Y^*)^U$. Here $w^*\text{-}\lim_{\alpha, U} y_\alpha^*$ is the w^* -limit of $(y_\alpha^*)_{\alpha \in \Lambda_0}$ through U , which exists by the Banach-Alaoglu Theorem.

Whenever $T \in B(X, Y)$, we can define an operator $\widehat{T} \in B(\widehat{Y}, \widehat{X})$ by sending $(y_\alpha^*)_{\alpha, U}$ to $(T^*y_\alpha^*)_{\alpha, U}$. Note that if $K \in K(X, Y)$ then $\widehat{K} = 0$, where $K(X, Y)$ denotes the space of all compact operators in $B(X, Y)$.

Theorem 2.1. *Suppose that X has the λ -BCAP. Then the operator $f : B(X)/K(X) \rightarrow B(\widehat{X})$, $\dot{T} \mapsto \widehat{T}$, is a norm one $(\lambda + 1)$ -embedding into $B(\widehat{X})$ satisfying*

$$f(\dot{I}) = I \text{ and } f(\dot{T}_1 \dot{T}_2) = f(\dot{T}_2) f(\dot{T}_1), \quad T_1, T_2 \in B(X).$$

Proof. It is easy to verify that f is a linear map, $f(\dot{I}) = I$, and $f(\dot{T}_1 \dot{T}_2) = f(\dot{T}_2) f(\dot{T}_1)$ for $T_1, T_2 \in B(X)$. If $T \in B(X)$, then clearly $\|f(\dot{T})\| \leq \|T\|$, and thus we also have $\|f(\dot{T})\| \leq \|T\|_e$. Hence $\|f\| \leq 1$. It remains to show that f is a $(\lambda + 1)$ -embedding (i.e., $\inf_{\|T\|_e > 1} \|f(\dot{T})\| \geq (\lambda + 1)^{-1}$).

To do this, let $T \in B(X)$ satisfy $\|T\|_e > 1$. Since X has the λ -BCAP, we can find a net of operators $(S_\alpha)_{\alpha \in \Lambda_0} \subset K(X)$ converging strongly to I such that $\sup_{\alpha \in \Lambda_0} \|S_\alpha\| \leq \lambda$. Then $\|T^*(I - S_\alpha)^*\| = \|(I - S_\alpha)T\| \geq \|T\|_e > 1$, $\alpha \in \Lambda_0$. Thus, there exists $(x_\alpha^*)_{\alpha \in \Lambda_0} \subset X^*$ such that $\|x_\alpha^*\| = 1$ and $\|T^*(I - S_\alpha)^*x_\alpha^*\| > 1$ for $\alpha \in \Lambda_0$.

Note that for every $x \in X$,

$$\limsup_{\alpha \in \Lambda_0} |\langle (I - S_\alpha)^*x_\alpha^*, x \rangle| = \limsup_{\alpha \in \Lambda_0} |\langle x_\alpha^*, (I - S_\alpha)x \rangle| \leq \limsup_{\alpha \in \Lambda_0} \|(I - S_\alpha)x\| = 0,$$

and so the net $((I - S_\alpha)^*x_\alpha^*)_{\alpha \in \Lambda_0}$ converges in the w^* -topology to 0. By the construction of U , this implies that

$$* \dots \dots \dots \sim * \dots \dots \dots$$

Immediately after the proof, the authors ask whether the theorem remains true without the BCAP hypothesis.

It follows that $\|f(\dot{T})\| \geq (1 + \lambda)^{-1}$ whenever $\|T\|_e > 1$. □

Remark 1. We do not know whether Theorem 2.1 is true without the hypothesis that X has the BCAP.

Remark 2. The embedding in Theorem 2.1 is an isometry if the approximating net can be chosen so that $\|I - S_\alpha\| = 1$ for every α . This is the case if, for example, the space X has a 1-unconditional basis. However, we do not know whether the embedding is an isometry if $X = L_p(0, 1)$ with $p \neq 2$.

If N is a subset of Y^* , then we can define a subset N' of $\widehat{Y}$ by

Result

Theorem 1. *Let X be an infinite-dimensional Banach space. In the notation of Boedihardjo–Johnson’s Calkin representation, if $T \in B(X)$ is noncompact, then $\widehat{T} \neq 0$. Consequently*

$$f : B(X)/K(X) \rightarrow B(\widehat{X}), \quad \dot{T} \mapsto \widehat{T},$$

is injective for every X , without assuming BCAP.

Idea of the Proof

The BCAP proof uses compact approximants S_α to produce weak-star-null functionals on which T^* stays large. For injectivity alone, compact approximants are unnecessary. If T is not compact, then $T(B_X)$ cannot be uniformly approximated by finite-dimensional subspaces. Thus there is a fixed positive lower bound for the quotient norms $\|Q_E T\|$, where $E \subset X$ ranges over finite-dimensional subspaces. By duality, for each E there is a norm-one functional in $E^\perp$ on which T^* is still large. These functionals form a weak-star-null element of the ultrapower $\widehat{X}$, because the index ultrafilter contains all tails $\{E : E \supset E_0\}$. Applying $\widehat{T}$ to this element gives a nonzero vector.

Proof. Let Λ_0 be the directed set of finite-dimensional subspaces of X , ordered by inclusion, and let $\mathcal{U}$ be an ultrafilter containing every tail

$$\{E \in \Lambda_0 : E \supset E_0\}.$$

For $E \in \Lambda_0$, let $Q_E : X \rightarrow X/E$ be the quotient map.

We first record a compactness criterion. If

$$\inf_{E \in \Lambda_0} \|Q_E T\| = 0,$$

then T is compact. Indeed, for every $\varepsilon > 0$ there is a finite-dimensional $E \subset X$ such that every point of $T(B_X)$ is within ε of E . Since $T(B_X)$ is bounded, it is within ε of a bounded subset of the finite-dimensional space E , and that bounded subset has a finite ε -net. Hence $T(B_X)$ is totally bounded. This is compactness of T .

Now assume T is not compact. Then

$$\eta := \inf_{E \in \Lambda_0} \|Q_E T\| > 0.$$

For every $E \in \Lambda_0$, duality gives

$$\|Q_E T\| = \|(Q_E T)^*\| = \|T^*|_{E^\perp}\|.$$

Choose $x_E^* \in E^\perp$ with $\|x_E^*\| = 1$ and

$$\|T^* x_E^*\| \geq \eta/2.$$

The bounded family $(x_E^*)_{E \in \Lambda_0}$ defines an ultrapower element

$$\xi = (x_E^*)_{E, \mathcal{U}} \in (X^*)^\mathcal{U}.$$

It lies in $\widehat{X}$. To see this, fix $x \in X$. On the tail $\{E : E \supset \text{span}\{x\}\}$, which belongs to $\mathcal{U}$, we have $x_E^*(x) = 0$. Hence the weak-star $\mathcal{U}$ -limit of x_E^* is zero. Moreover $\|\xi\| = 1$.

Applying $\widehat{T}$ gives

$$\widehat{T}\xi = (T^*x_E^*)_{E,\mathcal{M}}.$$

Therefore

$$\|\widehat{T}\xi\| = \lim_{E,\mathcal{M}} \|T^*x_E^*\| \geq \eta/2 > 0.$$

Thus $\widehat{T} \neq 0$. Since Boedihardjo–Johnson already observe that $\widehat{K} = 0$ for compact K , the induced map on $B(X)/K(X)$ is well-defined and injective. $\square$

What Remains Open

This proves only faithfulness. It does not give a constant $c > 0$ such that

$$\|\widehat{T}\| \geq c\|\dot{T}\|$$

for all $T \in B(X)$. Therefore it does not prove that f is an isomorphic embedding without BCAP in the sense of Theorem 2.1. It also does not settle the isometry question in Remark 2 for $L_p(0,1)$, $p \neq 2$.

Novelty and Literature Check

The run indexes were searched for arXiv:1301.0054, mean ergodic convergence, Calkin, BCAP, and Calkin representation phrases. Existing hits were adjacent Calkin-algebra problem packets, not this representation question. A bounded web search for exact phrases from Remark 1 and Theorem 2.1 did not find a later answer. Novelty confidence is moderate: this is a natural observation and may exist implicitly in operator-theory folklore, but it is not recorded in the run memory or surfaced by the search.

Human Review Recommendation

Reviewers should verify the compactness criterion and the dual identification $\|Q_ET\| = \|T^*|_{E^\perp}\|$. The rest of the argument is then a direct ultrapower-tail construction.

References

- [1] M. T. Boedihardjo and W. B. Johnson, *On Mean Ergodic Convergence in the Calkin Algebras*, arXiv:1301.0054.